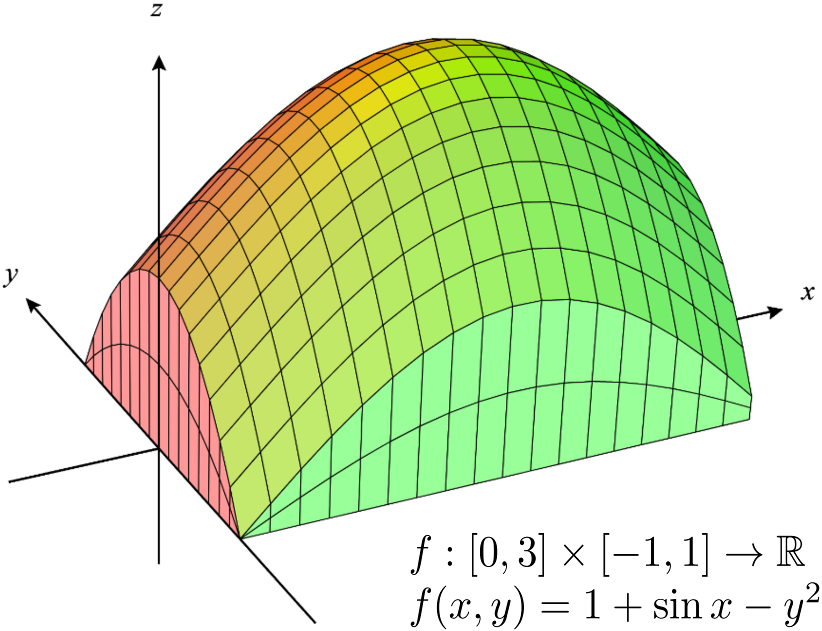
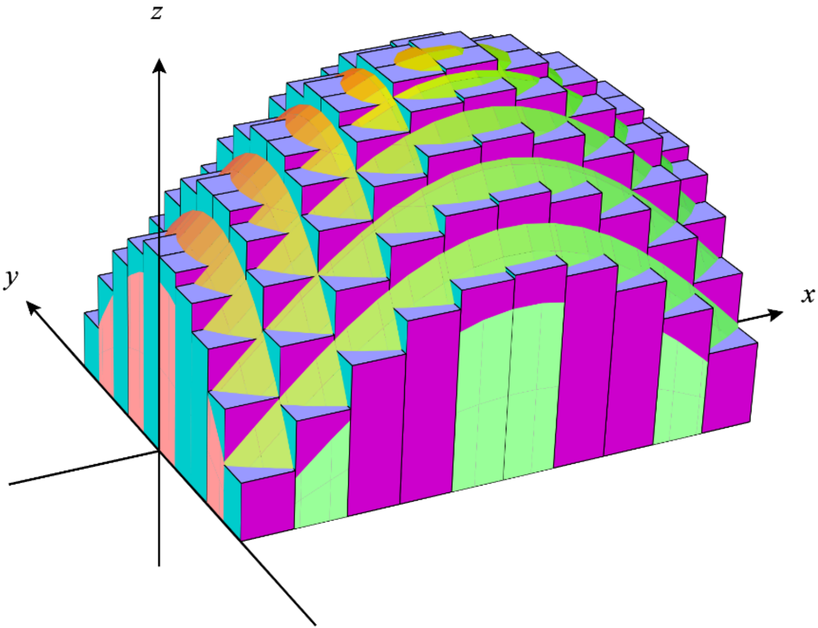
Mathematical Analysis 2

Seminar 12

Double integrals (1)



$$f : [0,3] \times [-1,1] \to \mathbb{R}$$
$$f(x,y) = 1 + \sin x - y^2$$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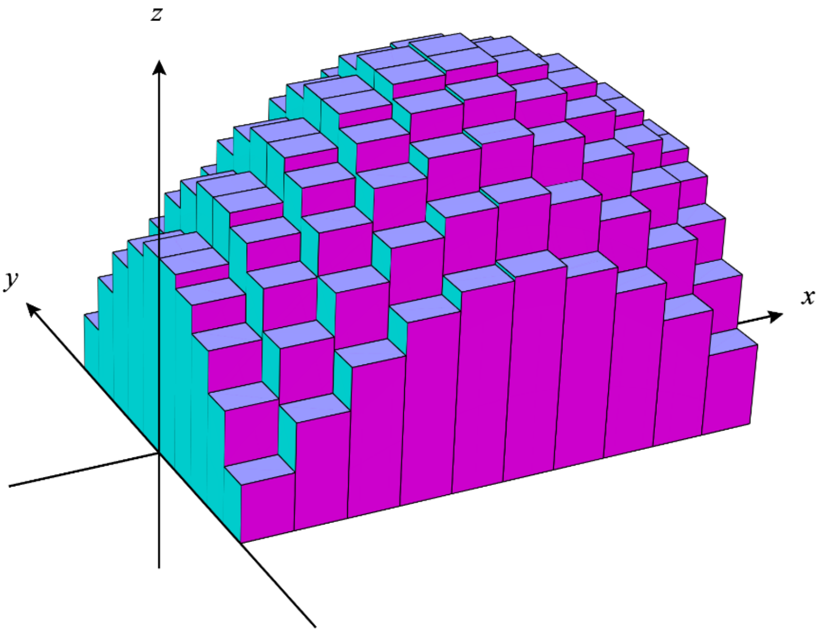
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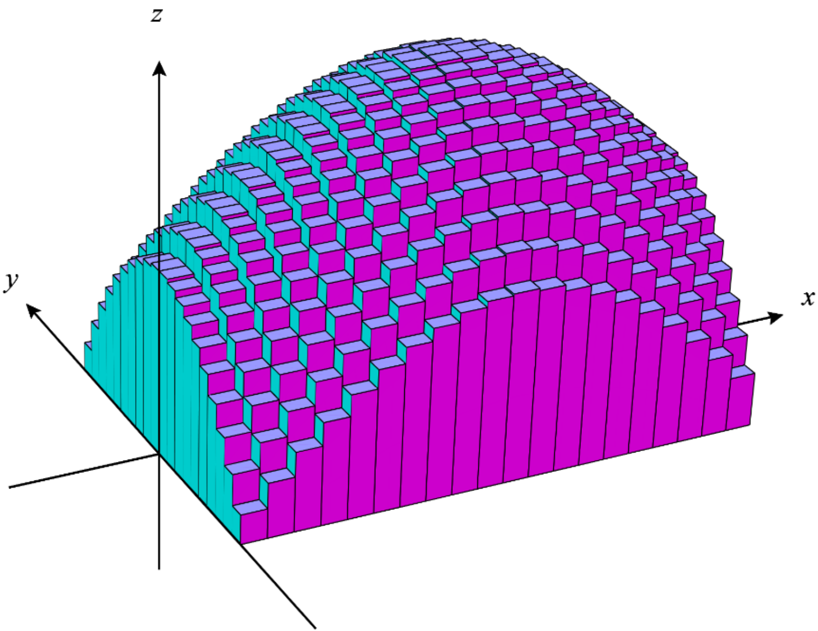
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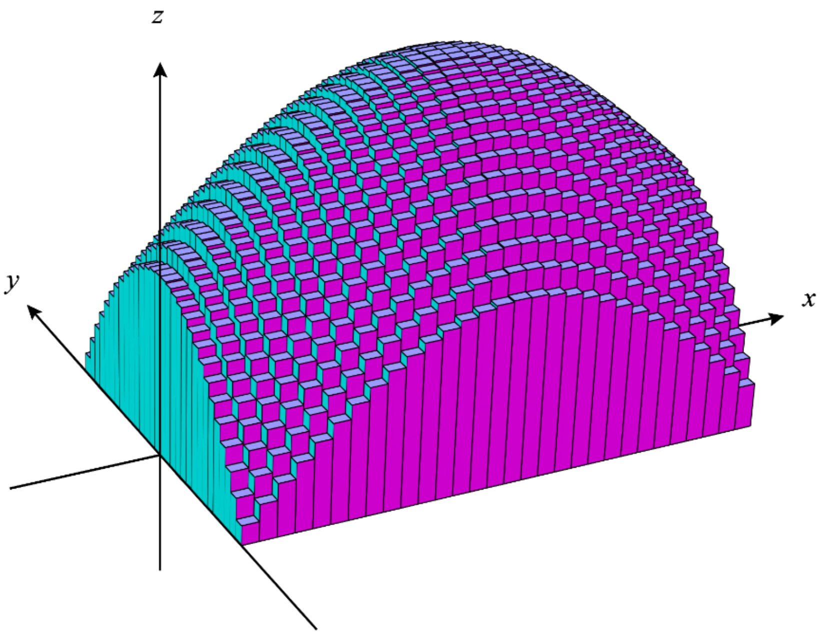
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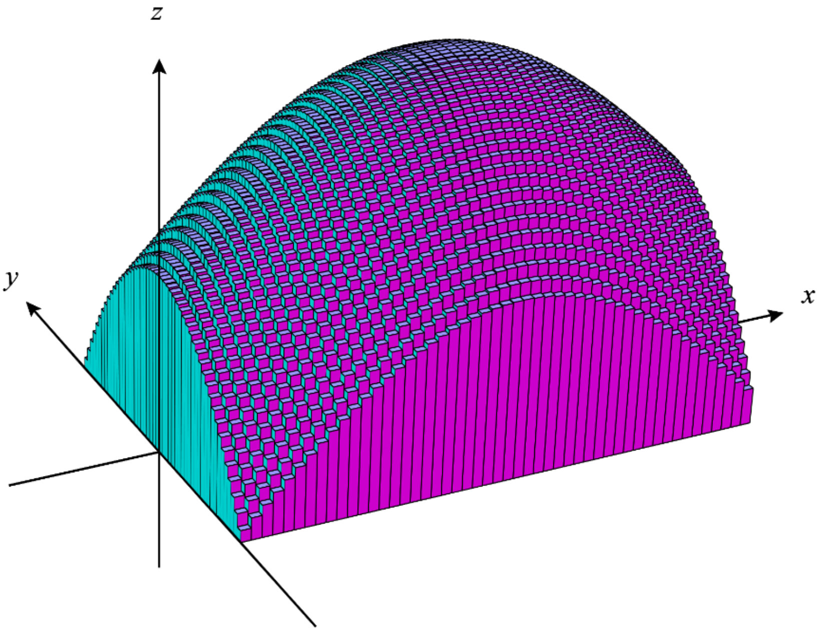
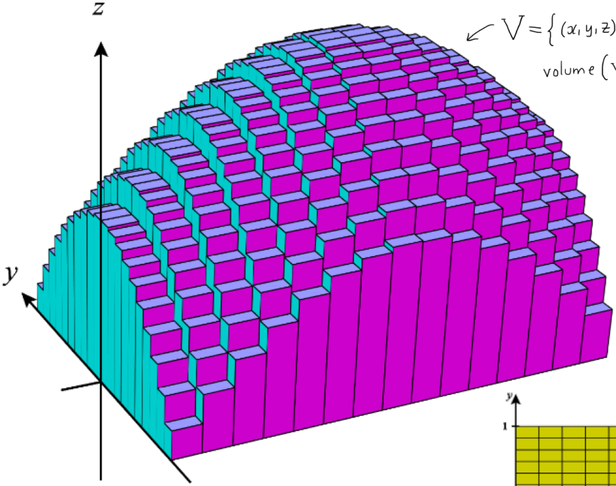
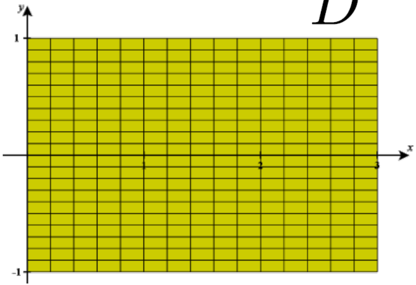

z
y
x

$$V = \{ (x,y,z) \in \mathbb{R}^3 \mid (x,y) \in D,\ 0 \leq z \leq f(x,y) \}$$

$$\text{volume}(V) = \iint_D f(x,y)\,dx\,dy$$



D



$$f : \overbrace{[0,3] \times [-1,1]}^{D} \to \mathbb{R}$$

$$f(x,y) = 1 + \sin x - y^2$$

Web links:

http://bit.ly/int_dubla_1S

http://bit.ly/int_dubla_1R

The double integral on rectangular domains

$D=[a,b]\times[c,d]$; $f: D\to\mathbb{R}$ continuous

$$\iint_D f(x,y)\,dx\,dy \overset{\text{notation}}{=\!=} \int_a^b\int_c^d f(x,y)\,dx\,dy \overset{\text{computation}}{=\!=} \begin{cases} \int_a^b\left(\int_c^d f(x,y)\,dy\right)dx \overset{\text{notation}}{=\!=} \int_a^b dx\int_c^d f(x,y)\,dy \\ \int_c^d\left(\int_a^b f(x,y)\,dx\right)dy \overset{\text{notation}}{=\!=} \int_c^d dy\int_a^b f(x,y)\,dx \end{cases}$$

interated integrals

Ex. 1 $$\iint_D \frac{y}{(1+x^2+y^2)^{\frac{3}{2}}}\,dx\,dy, \qquad D=[0,2]\times[1,2]$$

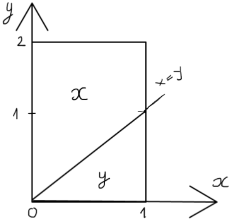
$$I=\int_0^2 dx\underbrace{\int_1^2\frac{y\,dy}{(1+x^2+y^2)^{3/2}}}_{J(x)} \quad ; \quad 1+x^2+y^2=u \,\big|\, d \qquad 2y\,dy=du$$

$$\Rightarrow J(x)=\int_{x^2+2}^{x^2+5}\frac{1}{2}u^{-\frac{3}{2}}du=\frac{1}{2}\cdot u^{-\frac{1}{2}}\cdot(-2)\Big|_{x^2+2}^{x^2+5}=$$

$$=\frac{1}{\sqrt{x^2+2}}-\frac{1}{\sqrt{x^2+5}} \Rightarrow I=\int_0^2\left(\frac{1}{\sqrt{x^2+2}}-\frac{1}{\sqrt{x^2+5}}\right)dx=\left[\ln\left(x+\sqrt{x^2+2}\right)-\ln\left(x+\sqrt{x^2+5}\right)\right]\Big|_0^2$$

$$=\left[\ln(2+\sqrt{6})-\ln\left(2+\sqrt{9}\right)\right]-\left[\ln\sqrt{2}-\ln\sqrt{5}\right]=\ln\frac{(2+\sqrt{6})\cdot\sqrt{5}}{5\cdot\sqrt{2}}=\ln\frac{\sqrt{2}+\sqrt{3}}{\sqrt{5}}.$$

Ex. 2 $\displaystyle\iint_D \min\{x,y\}\,dx\,dy, \qquad D=[0,1]\times[0,2]$



$$I=\int_0^1 dx\underbrace{\int_0^2 \min\{x,y\}\,dy}_{J(x)} \; ; \; J(x)=\int_0^x y\,dy+\int_x^2 x\,dy=\frac{y^2}{2}\Big|_{y=0}^{y=x}+xy\Big|_{y=x}^{y=2}$$

$$\Rightarrow J(x)=\frac{x^2}{2}+x(2-x)=-\frac{x^2}{2}+2x$$

$$I=\int_0^1\left(2x-\frac{x^2}{2}\right)dx=\left(x^2-\frac{x^3}{6}\right)\Big|_0^1=1-\frac{1}{6}=\frac{5}{6}.$$

? What if we choose $I=\int_0^2 dy\underbrace{\int_0^1 \min\{x,y\}\,dx}_{J(y)}$;

$$J(y)=\int_0^1 \min\{x,y\}\,dx, \quad y\in[0,2]$$

$$J(y)=\begin{cases} y\in[1,2]: \int_0^1 x\,dx=\frac{x^2}{2}\Big|_0^1=\frac{1}{2} \\ y\in[0,1]: \int_0^y x\,dx+\int_y^1 y\,dx=\frac{x^2}{2}\Big|_{x=0}^{x=y}+yx\Big|_{x=y}^{x=1} \end{cases}$$

$$=\frac{y^2}{2}+y(1-y)=y-\frac{y^2}{2}$$

$$J(y)=\begin{cases} y-\frac{y^2}{2}, & y\in[0,1] \\ \frac{1}{2}, & y\in[1,2] \end{cases}$$

$$I=\int_0^2 J(y)\,dy=\int_0^1\left(y-\frac{y^2}{2}\right)dy+\int_1^2\frac{1}{2}\,dy=\left(\frac{y^2}{2}-\frac{y^3}{6}\right)\Big|_0^1+\frac{1}{2}\cdot 1$$

$$=\frac{1}{2}-\frac{1}{6}+\frac{1}{2}=1-\frac{1}{6}=\frac{5}{6}.$$

Online visualization: http://bit.ly/int_dubla_02

Separation of variables:

Let $D = [a,b] \times [c,d]$. Then $\displaystyle\iint_D f(x)\cdot g(y)\,\mathrm{d}x\,\mathrm{d}y = \int_a^b \mathrm{d}x \int_c^d f(x) g(y)\,\mathrm{d}y =$

$$= \int_a^b \Big(f(x) \cdot \underbrace{\int_c^d g(y)\,\mathrm{d}y}_{\text{number}} \Big)\,\mathrm{d}x = \left(\int_a^b f(x)\,\mathrm{d}x\right)\left(\int_c^d g(y)\,\mathrm{d}y\right).$$

$\boxed{\textbf{Ex. 3}}$ $\displaystyle\iint_D e^{x+\sin y}\cdot \cos y\,\mathrm{d}x\,\mathrm{d}y, \qquad D = [0,1]\times\left[0,\frac{\pi}{2}\right]$

(Independent work)

The double integral on simple domains w.r.t. the axes of coordinates

(next time)